

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403249
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Cronenberg; Richard

Microinfuser with automatic needle retraction

Abstract

A device for delivering a fluid includes a housing defining an interior space and having a bottom surface configured for contacting a patient. The bottom surface defines a needle opening. The device also includes a reservoir disposed within the interior space of the housing for containing a fluid therein, a needle carrier disposed within the housing, and an injection needle supported by the needle carrier and defining a lumen. The lumen of the injection needle is configured to be placed in fluid communication with the reservoir. The injection needle is transitionable from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and a substantially shielded position in which the injection needle is disposed within the housing and the lumen of the needle is blocked by a portion of the housing.

Inventors:	Cronenberg; Richard (Mahwah, NJ)
Applicant:	Becton, Dickinson and Company (Franklin Lakes, NJ)
Family ID:	48170817
Assignee:	Becton, Dickinson and Company (Franklin Lakes, NJ)
Appl. No.:	17/718379
Filed:	April 12, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220241494 A1	Aug. 04, 2022

Related U.S. Application Data

continuation parent-doc US 16523778 20190726 US 11364336 child-doc US 17718379
continuation parent-doc US 14391666 US 10406280 20190910 WO PCT/US2013/035925

Publication Classification

Int. Cl.: A61M5/142 (20060101); A61M5/20 (20060101); A61M5/145 (20060101)

U.S. Cl.:

CPC A61M5/14248 (20130101); A61M5/20 (20130101); A61M5/2033 (20130101); A61M2005/14252 (20130101); A61M2005/14256 (20130101); A61M5/1454 (20130101); A61M2005/206 (20130101); A61M2205/582 (20130101); A61M2205/584 (20130101); A61M2205/585 (20130101)

Field of Classification Search

CPC: A61M (5/14248); A61M (5/20); A61M (5/2033); A61M (5/1454); A61M (2005/14252); A61M (2005/14256); A61M (2005/206); A61M (2205/582); A61M (2205/584); A61M (2205/585)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5312364	12/1993	Jacobs	N/A	N/A
5571133	12/1995	Yoon	N/A	N/A
7637891	12/2008	Wall	N/A	N/A
2001/0025168	12/2000	Gross et al.	N/A	N/A
2002/0123740	12/2001	Flaherty	N/A	N/A
2003/0105430	12/2002	Lavi	604/890.1	A61M 5/2033
2008/0097330	12/2007	King	N/A	N/A
2008/0281270	12/2007	Cross	604/122	A61M 5/158
2009/0130639	12/2008	Skinner	N/A	N/A
2010/0010454	12/2009	Marshall	N/A	N/A
2010/0145282	12/2009	Hansen	N/A	N/A
2010/0168683	12/2009	Cabiri	N/A	N/A
2011/0166512	12/2010	Both et al.	N/A	N/A
2011/0306929	12/2010	Levesque	604/150	A61M 5/14248
2012/0130207	12/2011	O'Dea	N/A	N/A
2013/0110049	12/2012	Cronenberg	604/239	A61M 5/1413
2016/0089056	12/2015	Limaye et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2010207762	12/2009	AU	N/A
1-130743	12/1988	JP	N/A
H04256757	12/1991	JP	N/A
2002505601	12/2001	JP	N/A
2004501721	12/2003	JP	N/A

2005538773	12/2004	JP	N/A
2004024211	12/2003	WO	N/A
2005018703	12/2004	WO	N/A
2005018705	12/2004	WO	N/A
2011075105	12/2010	WO	N/A
2011133823	12/2010	WO	N/A
2011146166	12/2010	WO	N/A
2012025639	12/2011	WO	N/A

Primary Examiner: Mehta; Bhisma

Assistant Examiner: Gonzalez; Lei

Attorney, Agent or Firm: Dickinson Wright PLLC

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/523,778, filed on Jul. 26, 2019, which is a continuation of U.S. patent application Ser. No. 14/391,666, filed on Oct. 9, 2014, now U.S. Pat. No. 10,406,280, which is the U.S. national stage of International Patent Application No. PCT/US2013/035925, filed on Apr. 10, 2013, which claims priority to U.S. Provisional Application No. 61/624,026, filed on Apr. 13, 2012, the disclosures of all of which are incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates generally to a device for delivering a fluid into the body of a patient by injection and, more particularly, to an infusing device with a mechanism for automatically retracting a needle after injection.

Description of Related Art

(2) Various types of automatic injection devices have been developed to allow drug solutions and other liquid therapeutic preparations to be administered by untrained personnel or to be self-injected. Generally, these devices include a reservoir that is pre-filled with the liquid therapeutic preparation, and some type of automatic needle-injection mechanism (usually of the spring-loaded type) that can be triggered by the user. When the volume of fluid or drug to be administered is generally below a certain volume, such as 1 mL, an auto injector is typically used, which typically has an injection time of about 10 to 15 seconds. When the volume of fluid or drug to be administered is above 1 mL, the injection time generally becomes longer resulting in difficulties for the patient to maintain contact between the device and the target area of the patient's skin. Further, as the volume of drug to be administered becomes larger, increasing the time period for injection becomes desirable. The traditional method for a drug to be injected slowly into a patient is to initiate an IV and inject the drug into the patient's body slowly. Such a procedure is typically performed in a hospital or outpatient setting.

(3) Certain devices allow for self-injection in a home setting and are capable of gradually infusing a liquid therapeutic preparation into the skin of a patient. In some cases, these devices are small enough (both in height and in overall size) to allow them to be “worn” by a patient while the liquid therapeutic preparation is being infused into the patient. These devices typically include a pump or other type of discharge mechanism to force the liquid therapeutic preparation to flow out of a reservoir and into the injection or infusion needle. Such devices also typically include a valve or

flow control mechanism to cause the liquid therapeutic preparation to begin to flow at the proper time and a triggering mechanism to initiate the injection. As with conventional syringes, needle-stick injuries with self-injection infusing devices are a concern.

SUMMARY OF THE INVENTION

(4) In one embodiment, a device for delivering a fluid includes a housing defining an interior space and having a bottom surface configured for contacting a patient. The bottom surface defines a needle opening. The device also includes a reservoir disposed within the interior space of the housing for containing a fluid therein, a needle carrier disposed within the housing, and an injection needle supported by the needle carrier and defining a lumen. The lumen of the injection needle is configured to be placed in fluid communication with the reservoir. The injection needle is transitionable from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and a substantially shielded position in which the injection needle is disposed within the housing and the lumen of the needle is blocked by a portion of the housing.

(5) The device may also include a needle actuator having a first ramp portion having an inclined section and a level section with the level section being generally parallel to the bottom surface of the housing. The needle carrier may include a first engagement for contacting the inclined section of the first ramp portion to transition the injection needle from the initial position to the use position. The needle actuator may also include a second ramp portion having an inclined section and a level section being generally parallel to the bottom surface of the housing and the needle carrier may include a second engagement for contacting the inclined section of the second ramp portion to transition the injection needle from the use position to the shielded position. The device may include a pad disposed within the housing, where the lumen of the injection needle is blocked by the pad when the injection needle is in the shielded position. The needle actuator may be configured to drive the injection needle into the pad when the injection needle is transitioned from the use position to the shielded position. At least a portion of the pad may be received by the needle actuator in the shielded position.

(6) The reservoir may be configured to automatically supply fluid to the injection needle after the injection needle is transitioned from the initial position to the use position. A needle actuator may be configured to engage the needle carrier to transition the injection needle from the initial position, to the use position, and the shielded position. The reservoir may comprise a syringe assembly including a syringe barrel for containing a fluid therein, a stopper disposed within the syringe barrel, and a plunger configured to advance the stopper within the syringe barrel to expel the fluid therefrom. The device may include a plunger stop engaged with the plunger when the injection needle is in the initial position, wherein the plunger stop is configured to engage the needle carrier and release the plunger stop from the plunger when the injection needle is transitioned from the initial position to the use position. The device may include a needle stop configured to restrict movement of the injection needle from the use position to the shielded position until a predetermined amount of fluid has been dispensed from the reservoir. The needle stop may include a rotatable body which engages the needle actuator and the plunger when the injection needle is in the use position with the needle stop disengaging from the plunger when the injection needle is transitioned from the use position to the shielded position.

(7) The device may also include an activation button, wherein deployment of the activation button allows the injection needle to transition from the initial position to the use position. The device may be provided with an indicator for indicating when the injection needle is in the initial position, the use position, and the shielded position.

(8) In a further embodiment, a device for delivering a fluid includes a housing defining an interior space and having a bottom surface configured for contacting a patient. The bottom surface defines a needle opening. The device further includes a reservoir disposed within the interior space of the housing for containing a fluid therein, a needle carrier disposed within the housing, and an injection

needle supported by the needle carrier and defining a lumen. The lumen of the injection needle is configured to be placed in fluid communication with the reservoir, wherein the injection needle is transitionable from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and a substantially shielded position in which the injection needle is disposed within the housing. The device also includes an indicator for indicating when the injection needle is in the initial position, the use position, and the shielded position.

(9) The housing may include at least one indicator lens, where the indicator is visible from an exterior of the housing via the indicator lens. The indicator may provide a visual and tactile indication of the position of the injection needle. The indicator may include an indicator post with the housing defining an indicator opening. The indicator post is disposed within the housing when the injection needle is in the initial position and the use position and extends through the indicator opening when the injection needle is in the shielded position.

(10) In another embodiment, a device for delivering a fluid includes a housing defining an interior space and having a bottom surface configured for contacting a patient. The bottom surface defines a needle opening. The device further includes a reservoir disposed within the interior space of the housing for containing a fluid therein, a needle carrier disposed within the housing, an activation member, and an injection needle supported by the needle carrier and defining a lumen. The lumen of the injection needle is configured to be placed in fluid communication with the reservoir. Deployment of the activation member causes the injection needle to transition from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and a substantially shielded position in which the injection needle is disposed within the housing.

(11) The device may include a needle actuator configured to engage the needle carrier to transition the injection needle from the initial position, to the use position, and the shielded position. The device may also include a needle stop configured to restrict movement of the injection needle from the use position to the shielded position until a predetermined amount of fluid has been dispensed from the reservoir. The activation member may engage the needle actuator when the injection needle is in the initial position. The needle stop may engage the needle actuator when the injection needle is in the use position, and the needle stop may be disengaged from the needle actuator when the injection needle is in the shielded position.

(12) Further details and advantages of the invention will become clear upon reading the following detailed description in conjunction with the accompanying drawing figures, wherein like parts are designated with like reference numerals throughout.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a device for delivering a fluid into a patient by injection in accordance with an embodiment of the present invention.

(2) FIG. 2 is a front view of the device of FIG. 1 in accordance with an embodiment of the present invention.

(3) FIG. 3 is a right view of the device of FIG. 1 in accordance with an embodiment of the present invention.

(4) FIG. 4 is a rear view of the device of FIG. 1 in accordance with an embodiment of the present invention.

(5) FIG. 5 is a top view of the device of FIG. 1 in accordance with an embodiment of the present invention.

(6) FIG. 6 is a left view of the device of FIG. 1 in accordance with an embodiment of the present invention.

invention.

(7) FIG. 7 is a bottom view of the device of FIG. 1 in accordance with an embodiment of the present invention.

(8) FIG. 8 is a bottom view of the device of FIG. 1 showing a needle cap removed in accordance with an embodiment of the present invention.

(9) FIG. 9 is a perspective view of the device of FIG. 1 showing a top cover removed in accordance with an embodiment of the present invention.

(10) FIG. 10 is a perspective view of the device of FIG. 1 showing detail of an activation button in accordance with an embodiment of the present invention.

(11) FIG. 11 is a perspective view of an activation button in accordance with an embodiment of the present invention.

(12) FIG. 12 is a bottom view of the device of FIG. 1 showing a bottom portion of the housing removed in accordance with an embodiment of the present invention.

(13) FIG. 13 is a perspective view of the device of FIG. 1 showing a first position of a plunger stop in accordance with an embodiment of the present invention.

(14) FIG. 14 is an enlarged perspective view of the device of FIG. 1 showing a second position of a plunger stop in accordance with an embodiment of the present invention.

(15) FIG. 15 is a partial perspective view of the device of FIG. 1 showing a second position of a plunger stop in accordance with an embodiment of the present invention.

(16) FIG. 16 is a cross-sectional view taken along line 16-16 of FIG. 10 in accordance with an embodiment of the present invention.

(17) FIG. 17 is a perspective view of a stopper in accordance with an embodiment of the present invention.

(18) FIG. 18 is a perspective view of a stopper in accordance with an embodiment of the present invention.

(19) FIG. 19 is a top view of the device of FIG. 1 showing engagement between a needle stop and needle actuator in accordance with an embodiment of the present invention.

(20) FIG. 20 is a partial bottom view of the device of FIG. 1 showing engagement between a needle stop and plunger in accordance with an embodiment of the present invention.

(21) FIG. 21 is a partial perspective view of the device of FIG. 1 showing a needle actuator removed and a plunger in an extended position in accordance with an embodiment of the present invention.

(22) FIG. 22 is a partial top view of the device of FIG. 1 showing a rotated position of a needle stop in accordance with an embodiment of the present invention.

(23) FIG. 23 is a top perspective view of the device of FIG. 1 showing a use position of an injector needle in accordance with an embodiment of the present invention.

(24) FIG. 24 is a bottom perspective view of the device of FIG. 1 showing a use position of an injector needle in accordance with an embodiment of the present invention.

(25) FIG. 25 is a perspective view of the device of FIG. 1 showing an injector needle transitioning from a use position to a shielded position in accordance with an embodiment of the present invention.

(26) FIG. 26 is a perspective view of the device of FIG. 1 showing an injector needle in a shielded position in accordance with an embodiment of the present invention.

(27) FIG. 27 is a partial perspective view of the device of FIG. 1 showing an injector needle in a shielded position with a needle actuator removed in accordance with an embodiment of the present invention.

(28) FIG. 28 is a bottom view of the device of FIG. 1 showing an injector needle in a shielded position in accordance with an embodiment of the present invention.

(29) FIG. 29 is a cross-sectional view of a needle actuator in accordance with an embodiment of the present invention.

(30) FIG. **30** is a schematic view of a needle carrier showing a pre-loaded position of the needle carrier just prior to movement of an injection needle to a use position in accordance with an embodiment of the present invention.

(31) FIG. **31** is a schematic view of a needle carrier showing a pre-loaded position of the needle carrier just prior to movement of an injection needle to a shielded position in accordance with an embodiment of the present invention.

(32) FIG. **32** is a top view of the device of FIG. **1** showing a needle carrier removed in accordance with an embodiment of the present invention.

(33) FIG. **33** is a perspective view of a device for delivering a fluid into a patient by injection in accordance with a further embodiment of the present invention.

(34) FIG. **34** is a perspective view of the device of FIG. **33** showing button lock in a second position in accordance with a further embodiment of the present invention.

(35) FIG. **35** is a perspective view of the device of FIG. **33** showing a top cover removed in accordance with a further embodiment of the present invention.

(36) FIG. **36** is a perspective view of the device of FIG. **33** showing the device at the end of an injection cycle in accordance with a further embodiment of the present invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

(37) For purposes of the description hereinafter, spatial orientation terms, if used, shall relate to the referenced embodiment as it is oriented in the accompanying drawing figures or otherwise described in the following detailed description. However, it is to be understood that the embodiments described hereinafter may assume many alternative variations and embodiments. It is also to be understood that the specific devices illustrated in the accompanying drawing figures and described herein are simply exemplary and should not be considered as limiting.

(38) Referring to FIGS. **1-16**, a device for delivering a fluid into a patient by injection **10** is shown. The device **10** generally includes a housing **12** having a first end **14** and a second end **16**. The housing **12** includes a bottom portion **18** and a top cover **20** that define an interior space **22**. The device further includes a reservoir **26**, an injection needle **28**, a needle carrier **30**, a needle actuator **32**, a plunger stop **34**, a needle stop **36**, and an activation button **38** that are accommodated by the housing **12**. The device **10** is utilized to inject a drug or medicament into a patient and is configured to be engaged with, such as mounted onto, a patient's skin for self-administration. Any form of medicament, e.g., liquid or slurry, including one or more pharmaceutically-active agents, may be administered by the device **10**.

(39) Referring to FIGS. **1-9**, the top cover **20** of the housing **12** includes a reservoir opening **44**, a button opening **46**, a first indicator lens **48**, a second indicator lens **50**, and third indicator lens **52**. The reservoir opening **44** allows for the visual inspection of the reservoir **26** without removing the top cover **20**. The button opening **46** receives the activation button **38** with a portion of the activation button **38** extending through the opening **46**. The first, second, and third indicator lenses **48**, **50**, **52** may allow visual inspection of a portion of the needle actuator **32** to provide an indication of the status of the device **10**. In particular, the first, second, and third indicator lenses **48**, **50**, **52** may be made from a transparent material to allow visual inspection of the interior space **22** of the housing **12**, as will be described herein. The bottom portion **18** of the housing **12** has a bottom surface **54** that defines a needle opening **56**. The bottom surface **54** is configured to be placed into contact with a target surface (not shown) of a patient. As shown in FIG. **7**, a needle cap **58** is received by the needle opening **56** in an initial pre-use position to shield the injection needle **28**, as shown in FIG. **8**, prior to use.

(40) Referring to FIGS. **10-12**, the activation button **38** includes a body **60** having a top surface **62** for engagement by a user. The top surface **62** of the body **60** extends through the button opening **46** of the housing **12**. A pair of tab members **64** extends outwardly from the body **60** and is received by the top cover **20** of the housing **12**. The body **60** of the activation button **38** includes a lock member **66** extending toward the needle actuator **32**. The activation button **38** is moveable within the

housing **12** and is configured to restrict movement of the needle actuator **32**. In particular, when the activation button **38** is in the position shown in FIG. **10**, the lock member **66** of the activation button **38** will be in interference engagement with the needle actuator **32** to restrict forward movement of the needle actuator **32**. Pressing downward on the activation button **38** will release the lock member **66** from the needle actuator **32** and start the operation of the device **10**, which will be discussed in more detail below.

(41) Referring to FIGS. **13-15**, the plunger stop **34** includes a body **70** having an attachment portion **72**, a plunger engagement portion **74**, and an actuator tab **76**. The attachment portion **72** is configured to be rotatably secured to the bottom portion **18** of the housing **12**. The plunger engagement portion **74** extends from the attachment portion **72** and is configured to engage the reservoir **26** as discussed in more detail below. The actuator tab **76** extends from the attachment portion **72** and includes an inclined planar surface **78** that is configured to engage a correspondingly shaped portion of the needle actuator **32**. The plunger stop **34** has a first position, as shown in FIG. **13**, and a second position, as shown in FIGS. **14** and **15**.

(42) Referring to FIGS. **13** and **16-18**, the reservoir **26** is embodied as a syringe assembly **82** having a syringe barrel **84**, a valve stopper **86**, a plunger **88**, and a fill stopper **90**, although other suitable reservoirs may be utilized. The syringe assembly **82** is configured to be disposed within the housing **12** and arranged along a longitudinal axis of the housing **12**. The syringe barrel **84** defines an interior space **92** for receiving a fluid therein and also receives the valve stopper **86** and fill stopper **90**. The plunger **88** includes a first end **94** and a second end **96** with the second end **96** of the plunger **88** engaged with the valve stopper **86**. A valve member **98** is received by the valve stopper **86** and is moveable between an open position and a closed position. The valve stopper **86** includes a passageway **102** having an inlet **104** and an outlet **106** generally extending through the valve stopper **86**. The passageway **102** of the valve stopper **86** is in fluid communication with the interior space **92** of the syringe barrel **84** when the valve member **98** is in the open position. A tube (not shown) extends from the outlet **106** of the passageway **102** to the injection needle **28**. The syringe assembly **82** further includes a plunger spring **108** configured to bias the valve stopper **86** towards the fill stopper **90** such that fluid within the syringe barrel **84** is displaced through the passageway **102** of the valve stopper **86**. A plunger guard **110** is disposed over the plunger spring **108**. The first end **94** of the plunger **88** defines a notch **112** configured to engage the plunger engagement portion **74** of the plunger stop **34**. The plunger **88** also includes an elongate arm **114** extending generally parallel to the syringe barrel **84** and configured to engage the needle stop **36** based on a predetermined position of the plunger **88** relative to the syringe barrel **84**. The plunger **88** is configured to move the valve stopper **86** toward the fill stopper **90** to dispense medicament or fluid from the syringe barrel **84**.

(43) Referring to FIGS. **6** and **7**, the syringe assembly **82** also includes a fill cover **116** that seals and encloses the fill stopper **90**. The fill cover **116** provides access to the fill stopper **90** during assembly of the device **10**. The fill cover **116** is configured to be secured to the housing **12** without being removable by a user of the device **10**.

(44) Referring to FIGS. **19-22**, the needle stop **36** includes a body **120** that defines a central opening **122** for receiving a pin **124** extending upward from the bottom portion **18** of the housing **12**. The body **120** of the needle stop **36** is rotatable about the pin **124** and includes an actuator surface **126** and a plunger surface **128**. The actuator surface **126** and the plunger surface **128** are configured to engage the needle actuator **32** and the plunger **88**, respectively. The plunger surface **128** is raised and positioned upwardly from the actuator surface **126**.

(45) Referring to FIGS. **23-29** and **32**, the needle actuator **32** includes an elongate body **132** with a first end **134** and a second end **136**. The body **132** of the needle actuator **32** includes a pair of sidewalls **138** extending between the first and second ends **134**, **136**. The body **132** of the actuator **32** defines an interior space **140** configured to receive a portion of the needle carrier **30**. First and second ramp portions **142**, **144** extend inward from one of the sidewalls **138** toward the interior

space **140**. The first ramp portion **142** includes a downward inclined section **146** and a level section **148**. The downward inclined section **146** of the first ramp portion **142** extends toward the bottom portion **18** of the housing **12** and towards the second end **136** of the needle actuator **32**. The level section **148** is generally parallel to the bottom surface **54** of the bottom portion **18** of the housing **12** and is positioned intermediate the ends **134**, **136** of the needle actuator **32**. The second ramp portion **144** includes an upward inclined section **150** extending away from the bottom portion **18** of the housing **12** and towards the second end **136** of the needle actuator **32**. The second ramp portion **144** also includes a level section **151** that is generally parallel to the bottom surface **54** of the bottom portion **18** of the housing **12**. Each of the sidewalls **138** of the needle actuator **32** includes the first and second ramp portions **142**, **144**, although only one of the sidewalls **138** may be provided with the ramp portions **142**, **144**. The needle actuator **32** also includes a plunger engagement tab **152** and an indicator tab **154**. The plunger engagement tab **152** includes an inclined planar portion **156** that is configured to engage the actuator tab **76** of the plunger stop **34**. The plunger engagement tab **152** is disposed adjacent to the first end **134** of the body **132** of the needle actuator **32**. The indicator tab **154** includes a planar surface **158** having first and second indicator colors **160**, **162**. The indicator tab **154** is disposed towards the first end **134** of the body **132** of the needle actuator **32** and is arranged substantially perpendicular to one of the sidewalls **138**.

(46) Referring to FIGS. **12**, **23-29**, and **32**, the needle actuator **32** further includes a button engagement **164** extending substantially perpendicular to one of the sidewalls **138** and positioned intermediate the first and second ends **134**, **136** of the needle actuator **32**. The button engagement **164** is configured to engage the lock member **66** of the activation button **38**. Upon downward movement of the activation button **38**, the button engagement **164** of the needle actuator **32** is configured to be released from engagement with the lock member **66** of the activation button **38**. As shown in FIG. **12**, a bottom **165** of the needle actuator **32** defines a bottom opening **166**. Further, the needle actuator **32** includes a needle spring **168** disposed within the interior space **140** of the needle actuator **32** and is configured to bias the needle actuator **32** forward toward the first end **14** of the housing **12**. The needle actuator **32** also includes a pad **170** disposed within the interior space **140** of the needle actuator **32** and positioned on a bottom surface **172** of the needle actuator **32**. The pad **170** is constructed from a rubber material, although other suitable materials may be utilized. Thus, the needle actuator **32** has a first position disposed adjacent to the second end **16** of the housing **12** and a second position disposed adjacent to the first end **14** of the housing **12** with the needle spring **168** configured to move the needle actuator **32** from the first position to the second position.

(47) Referring to FIGS. **9**, **10**, and **12**, the needle carrier **30** includes an elongate body **180** having a first end **182** and a second end **184**. The body **180** of the needle carrier **30** includes an attachment member **186** at the first end **182** of the body **180** that is configured to be secured to the bottom portion **18** of the housing **12**. The needle carrier **30** is arranged to be generally cantilevered from the attachment member **186** and extending from the first end **14** of the housing **12** towards the second end **16** of the housing **12**. A portion of the needle carrier **30** is configured to be received by and engaged with the needle actuator **32**. The needle carrier **30** also includes a first engagement **188** disposed between the ends **182**, **184** of the body **180** of the needle carrier **30** and a second engagement **190** disposed adjacent the second end **184** of the body **180** of the needle carrier **30**. The first and second engagements **188**, **190** extend radially outward from the body **180** and substantially perpendicular to a longitudinal axis of the needle carrier **30**. The needle carrier **30** further includes a generally cylindrical needle receiver **192** that receives and engages the injection needle **28**. The needle carrier **30** defines a passageway **194** having an inlet **196** that is configured to receive the tubing (not shown) from the outlet **106** of the valve stopper **86** of the syringe assembly **82**. A lumen (not shown) of the injection needle **28** is in fluid communication with the passageway **194** and the tubing from the valve stopper **86**. The needle carrier **30** is generally flexible or semi-rigid to allow the needle carrier **30** to flex from a non-biased state and return to the non-biased state

after flexing.

(48) Referring to FIGS. 9-32, upon assembly of the device **10**, the device **10** will be in an initial, pre-use state. As discussed in more detail below, the device **10** has a pre-use state, a use state, and a completed state. In the initial state, the syringe assembly **82** is received by the bottom portion **18** of the housing **12** and generally extends from the first end **14** of the housing **12** to the second end **16** of the housing **12**. The medicament is provided within the syringe barrel **84** and the plunger **88** is in a fully retracted position as shown in FIG. 9. The plunger spring **108** engages the bottom portion **18** of the housing **12** and the plunger **88** to bias the plunger **88** and valve stopper **86** towards the fill stopper **90**. The valve member **98** arranged within the valve stopper **86** is in a closed position. The attachment portion **72** of the plunger stop **34** is secured to the bottom portion **18** of the housing **12** and allows the plunger stop **34** to rotate relative to the bottom portion **18** of the housing **12**. The plunger engagement portion **74** of the plunger stop **34** engages the notch **112** of the plunger **88** to restrict movement of the plunger **88** towards the fill stopper **90**. The activation button **38** is in a raised position with the top surface **62** of the body **60** extending through the button opening **46** of the housing **12**. The lock member **66** is engaged with the button engagement **164** of the needle actuator **32** and prevents movement of the needle actuator **32**. The needle actuator **32** is positioned adjacent to the second end **16** of the housing **12** with the second end **136** of the needle actuator **32** abutting the second end **16** of the housing **12**. The needle spring **168** is disposed within the interior space **140** of the needle actuator **32** with a first end **200** of the spring **168** engaging the first end **134** of the needle actuator **32** and a second end **198** of the needle spring **168** engaging a spring seat **202** on the bottom portion **18** of the housing **12**. Thus, the needle actuator **32** is biased by the needle spring **168** and configured to move the needle actuator **32** from the second end **16** of the housing **12** to the first end **14** of the housing **12** upon release of the button engagement **164** with the lock member **66** of the activation button **38**.

(49) The syringe barrel **84** may be filled with a medicament by any suitable method. In one embodiment, with the fill cover **116** and fill stopper **90** removed from the housing **12**, the syringe barrel **84** is filled with medicament. A guide tube (not shown), which has a slightly smaller diameter than the inner diameter of the syringe barrel **84**, is used to squeeze and place the fill stopper **90** just above the medicament before retracting and leaving the fill stopper **90** in place. A further method uses a machine or apparatus to engage the top of the syringe barrel **84** and evacuate the syringe barrel **84** to a preset vacuum. When the preset vacuum is reached, the fill stopper **90** is pushed down into the syringe barrel **84** and the vacuum is released which causes atmospheric pressure to draw the fill stopper **90** into place while leaving no air gap. Another method of filling the syringe barrel **84** utilizes a piece of wire or string along the outside of the fill stopper **90** while the fill stopper **90** is positioned within the syringe barrel **84**. The wire or string creates a temporary break in the seal that allows air to escape from between the fill stopper **90** and the medicament as the fill stopper **90** is pushed into place.

(50) In the initial, pre-use state of the device **10**, the needle carrier **30** has a non-biased position with the elongate body **180** of the needle carrier **30** being substantially planar. The attachment member **186** of the needle carrier **30** is rotatably secured to a carrier seat **204** extending upwardly from the bottom portion **18** of the housing **12**. A portion of the needle carrier **30** is disposed within the needle actuator **32** with the elongate body **180** of the needle carrier **30** extending substantially parallel to the needle actuator **32**. The first engagement **188** of the needle carrier **30** is positioned adjacent to the first ramp portion **142** of the needle actuator **32**. The plunger engagement tab **152** of the needle actuator **32** is positioned adjacent to the actuator tab **76** of the plunger stop **34**. Further, the indicator tab **154** of the needle actuator **32** is positioned such that the first indicator color **160** is visible through the first indicator lens **48** of the housing **12** to indicate that the device **10** is ready to use.

(51) Referring to FIGS. 14 and 15, after removing the needle cap **58** from the needle opening **56** in the bottom portion **18** of the housing **12** and depressing the activation button **38** to move the

activation button **38** further within the housing **12**, the device **10** transitions from the pre-use position to the use position. After pressing the activation button **38**, the lock member **66** of the activation button **38** is released from the button engagement **164** of the needle actuator **32** to allow forward movement of the needle actuator **32** towards the first end **14** of the housing **12**. The needle actuator **32** moves forward, as shown in FIG. **14**, and the plunger engagement tab **152** of the needle actuator **32** engages the actuator tab **76** of the plunger stop **34** thereby causing the plunger stop **34** to rotate downward to the position shown in FIG. **15**. After the plunger stop **34** has rotated downward a sufficient distance, the plunger engagement portion **74** of the plunger stop **34** is released from the notch **112** of the plunger **88** thereby allowing the plunger **88** to be moved toward the fill stopper **90** by the plunger spring **108**. Upon movement of the plunger **88**, the valve member **98** within the valve stopper **86** is moved to the open position to place the injection needle **28** in fluid communication with the syringe barrel **84**. In particular, in the pre-use position, the plunger **88** is moveable within the valve stopper **86** a predetermined distance. When the plunger **88** disengages from the plunger stop **34**, the plunger **88** moves within the valve stopper **86** and causes the valve member **98** to move to the open position. Movement of the valve stopper **86** toward the fill stopper **90** causes fluid within the syringe barrel **84** to move through the passageway **102** in the valve stopper **86** through the tubing (not shown) and through the lumen of the injection needle **28**. (52) Referring to FIGS. **23** and **24**, when the device **10** is transitioned to the use state from the pre-use state, the injection needle **28** is moved to a use position where the injection needle **28** extends through the needle opening **56** in the bottom portion **18** of the housing **12**. In use, the bottom surface **54** of the bottom portion **18** of the housing **12** will be arranged against a target surface of a patient's body such that the injection needle **28** pierces the target surface of the patient. As discussed above, when the injection needle **28** is moved from the initial position to the use position, fluid within the syringe barrel **84** is automatically supplied to the lumen of the injection needle **28** for infusion into the patient.

(53) Referring to FIGS. **29** and **30**, during the transition of the injection needle **28** from the initial position to the use position, the needle carrier **30** is pre-loaded to snap the needle carrier **30** downward to quickly pierce the target area of a patient's body with the injection needle **28**. In particular, as the needle actuator **32** moves forward toward the first end **14** of the housing **12**, the first engagement **188** of the needle carrier **30** engages the bottom of the inclined section **146** of the first ramp portion **142** while the second engagement **190** of the needle carrier **30** engages the top of the level section **148** of the first ramp portion **142**. Further movement of the needle actuator **32** causes the intermediate portion between the first and second ends **182**, **184** of the needle carrier **30** to flex, as shown in FIG. **30**, while the second end **184** of the needle carrier **30** is supported by the level section **148** of the first ramp portion **142**. Continued movement of the needle actuator **32** causes the second engagement **190** of the needle carrier **30** to extend beyond the level section **148** of the first ramp portion **142** thereby releasing the second end **184** of the needle carrier **30** and causing the needle carrier **30** and injection needle **28** to snap downwardly.

(54) Referring to FIG. **23**, when the device **10** is in the use state and the injection needle **28** is in the use position, the needle carrier **30** is positioned between the first and second ends **14**, **16** of the housing **12**. The first indicator color **160** of the indicator tab **154** is positioned below the second indicator lens **50** of the housing **12** to indicate that the device **10** is in the use state.

(55) Referring to FIGS. **19** and **20**, continued movement of the needle actuator **32** is restricted when the device **10** is in the use state until a predetermined amount of medicament has been dispensed from the syringe barrel **84**. In particular, the actuator surface **126** of the needle stop **36** engages the needle actuator **32** and the plunger surface **128** engages the elongate arm **114** of the plunger **88**. The needle stop **36** is restricted from rotating when the needle actuator **32** engages the needle stop **36** due to the interference of the plunger surface **128** of the needle stop **36** with the elongate arm **114** of the plunger **88**. Upon further movement of the plunger **88**, such as when the valve stopper **86** engages the fill stopper **90**, the plunger surface **128** of the needle stop **36** will be

free from interference with the elongate arm **114** of the plunger **88** to allow the needle actuator **32** to rotate the needle stop **36** and continue movement of the needle actuator **32** toward the first end **14** of the housing **12**.

(56) Referring to FIGS. **21**, **22**, and **25-28**, as the needle actuator **32** is moved past the needle stop **36** toward the first end **14** of the housing **12**, the device **10** is transitioned from the use state to the completed state. As discussed above, after the plunger **88** moves the valve stopper **86** into engagement with the fill stopper **90** after a predetermined amount of time, the needle stop **36** rotates and allows the needle actuator **32** to move to the first end **14** of the housing **12** with the first end **134** of the needle actuator **32** abutting the first end **14** of the housing **12**, as shown in FIG. **26**. The rotation of the needle stop **36** and the clearing of the elongate arm **114** of the plunger **88** is shown in FIGS. **21** and **22**. The injection needle **28** is transitioned from the use position to the shielded position where the lumen of the injection needle **28** is blocked and where the injection needle **28** is positioned within the housing **12**. In particular, in the shielded position, the injection needle **28** engages the pad **170** positioned within the needle actuator **32** to block the lumen of the injection needle **28** and prevent fluid from exiting the injection needle **28**. In the completed state of the device, the second indicator color **162** of the indicator tab **154** of the needle actuator **32** is positioned below the third indicator lens **52** of the housing **12** to indicate the completed state of the device **10**. Further, as shown in FIG. **28**, the bottom **165** of the needle actuator **32** closes the needle opening **56** in the bottom portion **18** of the housing **12**.

(57) Referring to FIGS. **25**, **29**, and **31**, during the transition of the injection needle **28** from the use position to the shielded position, the needle carrier **30** is pre-loaded to snap the needle carrier **30** downward to engage the pad **170** positioned within the needle actuator **32**. As the needle actuator **32** moves forward toward the first end **14** of the housing **12**, the first engagement **188** of the needle carrier **30** engages the bottom of the level section **148** of the first ramp portion **142** while the second engagement **190** of the needle carrier **30** engages the top of the inclined section **150** of the second ramp portion **144**. Further movement of the needle actuator **32** causes the second end **184** of the needle carrier **30** to flex upwardly as the second engagement **190** proceeds along the inclined section **150** of the second ramp portion **144** while the first engagement **188** slides along the level section **148** of the first ramp portion **142**. Continued movement of the needle actuator **32** towards the first end **14** of the housing **12** causes the second engagement **190** of the needle carrier **30** to extend beyond the level section **151** of the second ramp portion **144** thereby causing the needle carrier **30** and injection needle **28** to snap downwardly into the pad **170**. After the injection needle **28** engages the pad **170**, the lumen of the injection needle **28** is blocked by the pad **170** to prevent fluid from leaking out of the device **10** when the device **10** is in the completed state.

(58) Referring to FIGS. **33-36**, a second embodiment of a device **220** for delivering a fluid into a patient by injection is shown. The device **220** is similar to the device **10** shown in FIGS. **1-32**, except for the differences discussed below. Like reference numbers are used for like elements. Instead of providing the first, second, and third indicator lenses **48**, **50**, **52**, the device **220** shown in FIGS. **33-36** includes an indicator lens **222** that extends outward from the top cover **20** of the housing **12**. Further, the needle actuator **32** includes a raised indicator tab **224** that extends upwardly from the body **132** of the needle actuator **32**. The raised indicator tab **224** of the needle actuator **32** extends into the indicator lens **222** and is received by the indicator lens **222**. The needle actuator **32** also includes an indicator post **226** that extends outward from the first end **134** of the needle actuator **32**. The first end **14** of the housing **12** defines an indicator opening **228** configured to receive the indicator post **226**. As the device **220** is moved from the pre-use state, the use state, and the completed state, the indicator tab **224** on the needle actuator **32** will move within the indicator lens **222** to provide an indication of the state of the device **220**. Further, when the device **220** is in the completed state, the indicator post **226** of the needle actuator **32** will protrude through the indicator opening **228** of the housing **12** to provide visual and tactile feedback of the completed state of the device **220**. The indicator tab **224** and the indicator lens **222** arrangement of the present

embodiment allows the indicator tab **224** to be viewed from a larger viewing area compared to the indicator arrangement of the first embodiment.

(59) Referring again to FIGS. **33-36**, the device **220** of the second embodiment also includes a two piece activation button **230** having a first portion **232** and a second portion **234**. The first portion **232** is moveable relative to the second portion **234** and has a locked position (shown in FIG. **34**) and a use position (shown in FIG. **33**). When the first portion **232** is in the locked position, a protrusion **236** extends radially outward to interfere with the button opening **46** in the top cover **20** of the housing **12** to prevent the activation button **38** from being depressed. When the first portion **232** of the activation button **230** is moved to the use position, the protrusion **236** is received by the second portion **234** and allows the activation button **230** to be depressed to begin the injection process. The activation button **230**, however, otherwise operates in a similar manner as described above in connection with the first embodiment. Accordingly, the two-piece activation button **230** allows the activation button to be locked to prevent accidental operation of the device.

(60) While several embodiments were described in the foregoing detailed description, those skilled in the art may make modifications and alterations to these embodiments without departing from the scope and spirit of the invention. Accordingly, the foregoing description is intended to be illustrative rather than restrictive. The invention described hereinabove is defined by the appended claims and all changes to the invention that fall within the meaning and the range of equivalency of the claims are embraced within their scope.

Claims

1. A device for delivering a fluid comprising: a housing defining an interior space and having a bottom surface configured for contacting a patient, the bottom surface defining a needle opening; a reservoir disposed within the interior space of the housing for containing a fluid therein; a biased plunger configured to move within the reservoir; a needle carrier disposed within the housing; an injection needle supported by the needle carrier and defining a lumen in selective fluid communication with the reservoir; a biased needle actuator movably disposed within the housing, wherein via interaction between the needle actuator and the needle carrier, movement of the needle actuator controls movement of the injection needle from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and subsequently to a substantially shielded position; and a needle stop, wherein when the injection needle reaches the use position, in a first position, the needle stop contacts the plunger and engages the needle actuator to prevent further movement of the needle actuator; wherein upon a predetermined displacement of the plunger, the needle stop moves from the first position to a second position, permitting movement of the needle actuator, to move the needle to the substantially shielded position.
2. The device according to claim 1, wherein the plunger and the needle actuator move substantially parallel to the bottom surface.
3. The device according to claim 1, wherein the plunger and the needle actuator move substantially parallel to each other.
4. The device according to claim 1, wherein the plunger and the needle actuator move in the same direction.
5. The device according to claim 1, wherein the needle stop rotates between the first and second positions.
6. The device according to claim 1, further comprising an indicator tab disposed on and movable with the needle actuator for separately indicating when the injection needle is in the initial position, the use position, and the shielded position.
7. A device for delivering a fluid comprising: a housing defining an interior space and having a bottom surface configured for contacting a patient, the bottom surface defining a needle opening; a

reservoir disposed within the interior space of the housing for containing a fluid therein; a plunger configured to move within the reservoir; a first biasing member biasing the plunger; a needle carrier disposed within the housing; an injection needle supported by the needle carrier and defining a lumen in selective fluid communication with the reservoir; and a needle actuator movably disposed within the housing; a second biasing member biasing the needle actuator; wherein via interaction between the needle actuator and the needle carrier, movement of the needle actuator controls movement of the injection needle from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and subsequently to a substantially shielded position; and a needle stop, wherein when the injection needle reaches the use position, in a first position, the needle stop engages the plunger and engages the needle actuator to prevent further movement of the needle actuator; wherein upon a predetermined displacement of the plunger, the needle stop moves from the first position to a second position, permitting movement of the needle actuator, to move the needle to the substantially shielded position.

8. The device according to claim 7, wherein the plunger and the needle actuator move substantially parallel to the bottom surface.

9. The device according to claim 7, wherein the plunger and the needle actuator move substantially parallel to each other.

10. The device according to claim 7, wherein the plunger and the needle actuator move in the same direction.

11. The device according to claim 7, wherein the needle stop rotates between the first and second positions.

12. The device according to claim 7, further comprising an indicator tab disposed on and movable with the needle actuator for separately indicating when the injection needle is in the initial position, the use position, and the shielded position.

13. A device for delivering a fluid comprising: a housing defining an interior space and having a bottom surface configured for contacting a patient, the bottom surface defining a needle opening; a reservoir disposed within the interior space of the housing for containing a fluid therein; a plunger configured to move within the reservoir; a first biasing member biasing the plunger; a needle carrier disposed within the housing; an injection needle supported by the needle carrier and defining a lumen in selective fluid communication with the reservoir; and a needle actuator movably disposed within the housing; a second biasing member biasing the needle actuator; wherein via interaction between the needle actuator and the needle carrier, movement of the needle actuator controls movement of the injection needle from an initial position in which the injection needle is disposed within the housing, to a use position in which the injection needle extends through the needle opening, and subsequently to a substantially shielded position; and a needle stop having a plunger surface and an actuator surface; wherein when the injection needle reaches the use position, the actuator surface of the needle stop engages the needle actuator to prevent further movement of the needle actuator, and the plunger surface slidingly engages a surface of the plunger, preventing movement of the needle stop as the plunger displaces to eject fluid from the reservoir; and wherein upon a predetermined displacement of the plunger, the surface of the plunger bypasses the plunger surface of the needle stop, permitting movement of the needle stop by the needle actuator, to disengage the actuator surface of the needle stop from the actuator, and permit further movement of the needle actuator, to move the needle to the substantially shielded position.

14. The device according to claim 13, wherein the plunger and the needle actuator move substantially parallel to the bottom surface.

15. The device according to claim 13, wherein the plunger and the needle actuator move substantially parallel to each other.

16. The device according to claim 13, wherein the plunger and the needle actuator move in the

same direction.

17. The device according to claim 13, wherein the needle stop rotates between the first and second positions.

18. The device according to claim 13, further comprising an indicator tab disposed on and movable with the needle actuator for separately indicating when the injection needle is in the initial position, the use position, and the shielded position.
